

Government Polytechnic, Osmanabad.

Second Progressive Assessment Test of Theory AY-2023-24

Program Name: Computer Engineering Program (CO)

Semester : Third (CO3I)

Course Title : Data Structure using C - 22317

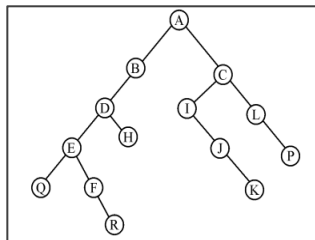
Marks : 20

Time: 1 Hour

Date : 01 / 11 /2023

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.



CO N	Course Outcome(CO)
a	Perform Basic Operations on Array
b	Apply different searching and sorting techniques.
c	Implement basic operations on stack and queue using Array representation.
d	Implement Basic Operations on Linked List.
e	Implement Program to create and traverse tree to solve problems.

Q.1 Attempt any FOUR.

08 Marks

QN	Question	Marks	CO
a	Define Binary tree and Binary search tree?	2	e
b	Define Queue and enlist applications of Queue?	2	c
c	Define term w.r.t Binary Tree 1) In-degree 2) Out-degree?	2	e
d	With Diagram, Explain terms: node, next pointer?	2	d
e	Convert the following infix expression to its postfix form using stack $A + B - C * D / E + F$	2	c
f	Explain Circular Linked List with diagram?	2	d

Q.2 Attempt any THREE.

12 Marks

QN	Question	Marks	CO
a	Write an algorithm to traverse a linked list?	4	d
b	For binary tree given above, write in-order, pre-order and post-order traversal?	4	e
c	Explain Linear Queue overflow and underflow conditions with examples?	4	c
d	With Diagram, explain procedure for deletion of a node from Linked list 1) At beginning 2) At Middle	4	d

Government Polytechnic, Osmanabad.

Second Progressive Assessment Test of Theory AY-2023-24

Program Name: Computer Engineering Program (CO)

Semester : Third (CO3I)

Course Title : Data Structure using C - 22317

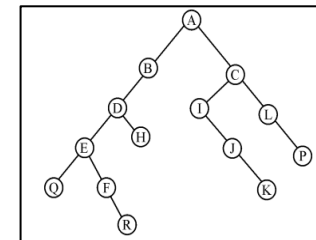
Marks : 20

Time: 1 Hour

Date : 01 / 11 /2023

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.



CO N	Course Outcome(CO)
a	Perform Basic Operations on Array
b	Apply different searching and sorting techniques.
c	Implement basic operations on stack and queue using Array representation.
d	Implement Basic Operations on Linked List.
e	Implement Program to create and traverse tree to solve problems.

Q.1 Attempt any FOUR.

08 Marks

QN	Question	Marks	CO
a	Define Binary tree and Binary search tree?	2	e
b	Define Queue and enlist applications of Queue?	2	c
c	Define term w.r.t Binary Tree 1) In-degree 2) Out-degree?	2	e
d	With Diagram, Explain terms: node, next pointer?	2	d
e	Convert the following infix expression to its postfix form using stack $A + B - C * D / E + F$	2	c
f	Explain Circular Linked List with diagram?	2	d

Q.2 Attempt any THREE.

12 Marks

QN	Question	Marks	CO
a	Write an algorithm to traverse a linked list?	4	d
b	For binary tree given above, write in-order, pre-order and post-order traversal?	4	e
c	Explain Linear Queue overflow and underflow conditions with examples?	4	c
d	With Diagram, explain procedure for deletion of a node from Linked list 1) At beginning 2) At Middle	4	d